

Practice 01

Note: Do not use arrays or nested loop in any task:

Q1: Write a program to input 15 float numbers and print the following:

§ Average

§ Maximum and 2nd Maximum

§ Minimum and 2nd Minimum

Sample case:

Enter 15 numbers: 10.5 56.2 30.9 45.62 15.5 9.7 89.6 19.74 25.08 65.03 73.42 3.62
55.25
99.99 78.92
Average: 45.27
Maximum Number: 99.9
2nd Maximum Number: 89.6
Minimum Number: 3.62
2nd Minimum Number: 9.7

Q2: Write a program to input 4 integer numbers from user and sort them in ascending order.

Sample case:

Enter 4 numbers: 15 90 35 74

Numbers in ascending order: 15 35 74 90

Q3: Write a program:

§ Input the starting and ending numbers from user

§ Input a number k

§ Print all the numbers divisible by k in range from start to end in a single line

Sample case:

Enter starting number: 10

Enter ending number: 20

Enter k: 3

Numbers divisible by k: 12 15 18

Q4: Input ten numbers. For every number except first number check if the number is larger than all

previous numbers, print "this number is larger than X", where X is previous largest number. At the end

print the kth number was largest.

Sample case:

Enter number: 10

Enter number: 5

Enter number: 25

25 is larger than 10

Enter number: 14

Enter number: 79

79 is larger than 25

Enter number: 62

Enter number: 46

Enter number: 81

81 is larger than 79

Enter number: 15

Enter number: 12

Enter number: 3

81 is the largest number

Q5: Write a program:

§ Input the number of sides (0, 3, 4)

§ If number of sides is 0, input radius and calculate area of circle

§ If number of sides are 3, input the height and width to calculate the area of triangle

§ If number of sides are 4, input the height and width, first check if it is a square or a rectangle and then calculate area

§ Print the name of the shape and the area.

Sample case:

Enter number of sides: 0

Enter radius: 5

A circle, The area is: 78.55

Q6: Write a program to convert the given number into octal form.

Hint: Integer division and remainder by 8

Sample case:

Enter a number: 100

Octal form: 144 ($100 / 8 = 12$, $100 \% 8 = 4$, next $12 / 8 = 1$, $12 \% 8 = 4$, next $1 \% 8 = 1$ & $1 / 8 = 0$ to stop the loop)

Q7: Write a program to calculate nth Fibonacci number. Fibonacci sequence is a series of numbers (0, 1, 1, 2, 3, 5, 8,) where each number is the sum of the two preceding ones, starting from 0 and 1. Fibonacci of 0 is 0 and 1 is 1.

Sample case:

Enter n: 5

Nth Fibonacci number: 3